

projectB: Exploratory Biomarker Analysis

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2026/07/21

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Dataset Overview

```
df <- readRDS("data/processed_data.rds")
glimpse(df)
```

```
## Rows: 30
## Columns: 6
## $ sample_id      <chr> "S001", "S002", "S003", "S004", "S005", "S006", "S007"~
## $ biomarker_ng_ml <dbl> 8.9, 10.1, 9.4, 11.0, 9.8, 10.5, 11.7, 8.5, 10.8, 9.7,~
## $ group          <fct> Control, Control, Control, Control, Control, Control, ~
## $ sex            <fct> F, F, F, M, F, F, M, F, M, F, M, F, M, M, F, M, F, ~
## $ age            <int> 26, 31, 34, 38, 42, 45, 48, 29, 51, 37, 55, 43, 57, 60~
## $ batch          <fct> B1, B1, B2, B1, B2, B1, B2, B1, B2, B1, B2, B1, B2, B1~
```

Exploratory Questions

1. How are the biomarker measurements distributed?

Biomarker concentrations range from 8.5 ng/mL to 18.1 ng/mL with an overall mean of approximately 13.24 ng/mL.

```
knitr::include_graphics("output/figures/fig1_biomarker_distribution.png")
```

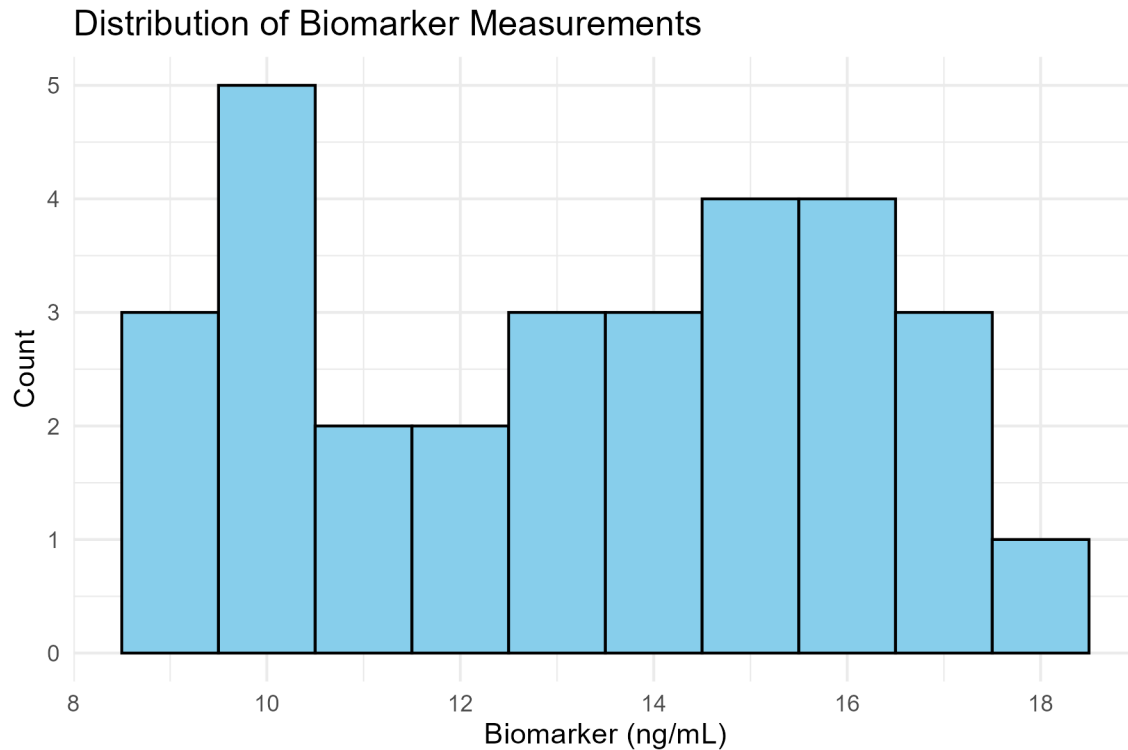


Figure 1: Distribution of Biomarker Concentration

2. Does the treatment group appear different from the control group?

YES. The Control group has a mean level of **10.87 ng/mL**, whereas the Treatment group has a mean level of **15.61 ng/mL**.

```
knitr::include_graphics("output/figures/fig2_group_comparison.png")
```

3. Does age appear to influence the biomarker?

YES. There is a positive correlation between age and biomarker concentration in both study arms.

```
knitr::include_graphics("output/figures/fig3_age_vs_biomarker.png")
```

4. Does sex appear to influence the biomarker?

YES, males display higher than females, but driven by AGE.

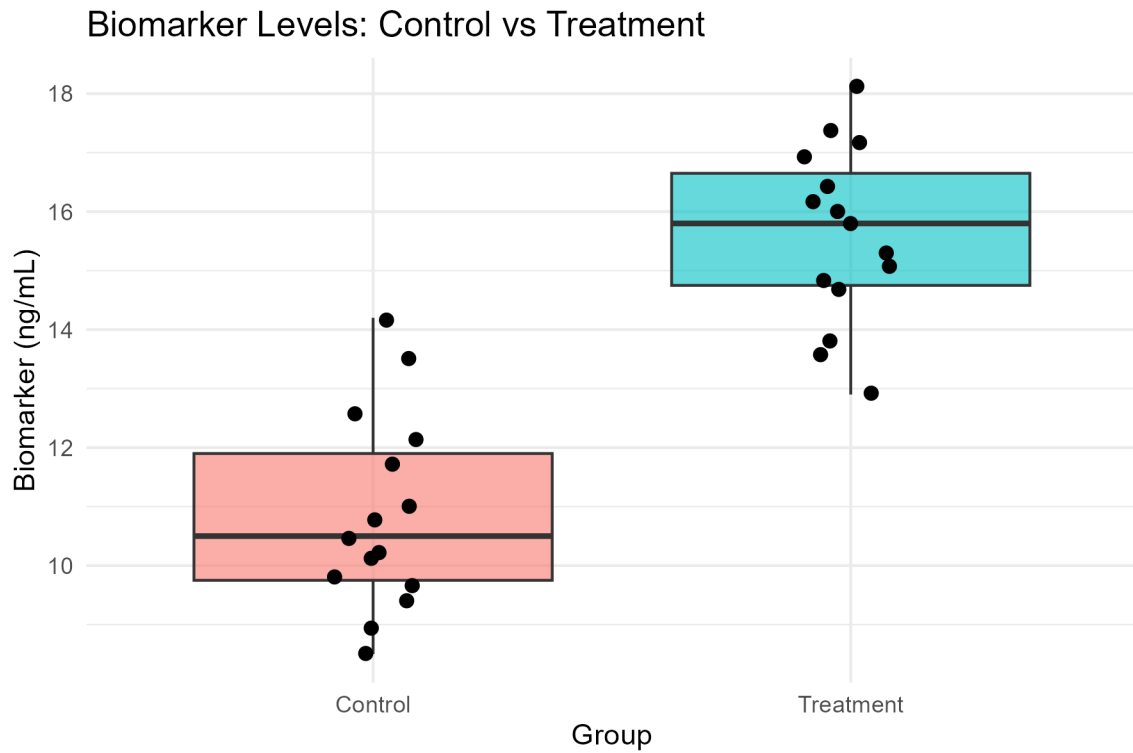


Figure 2: Control vs Treatment Biomarker Comparison

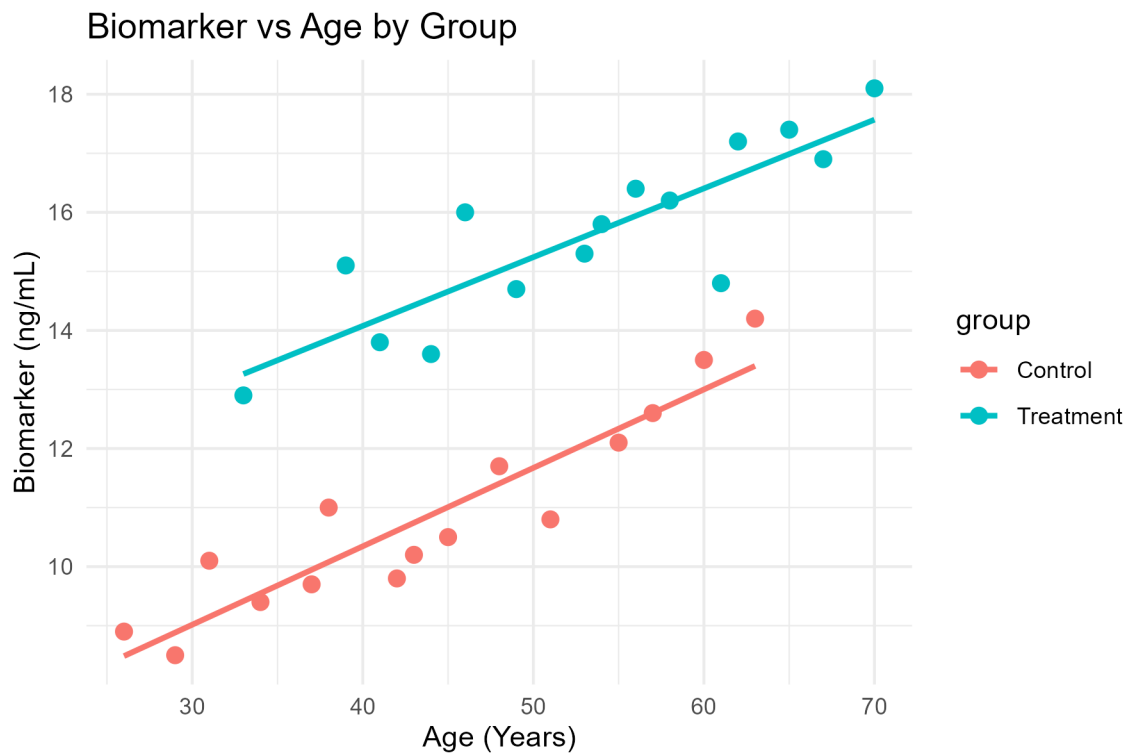


Figure 3: Biomarker vs Age Stratified by Group

```
knitr::include_graphics("output/figures/fig4_sex_vs_biomarker.png")
```

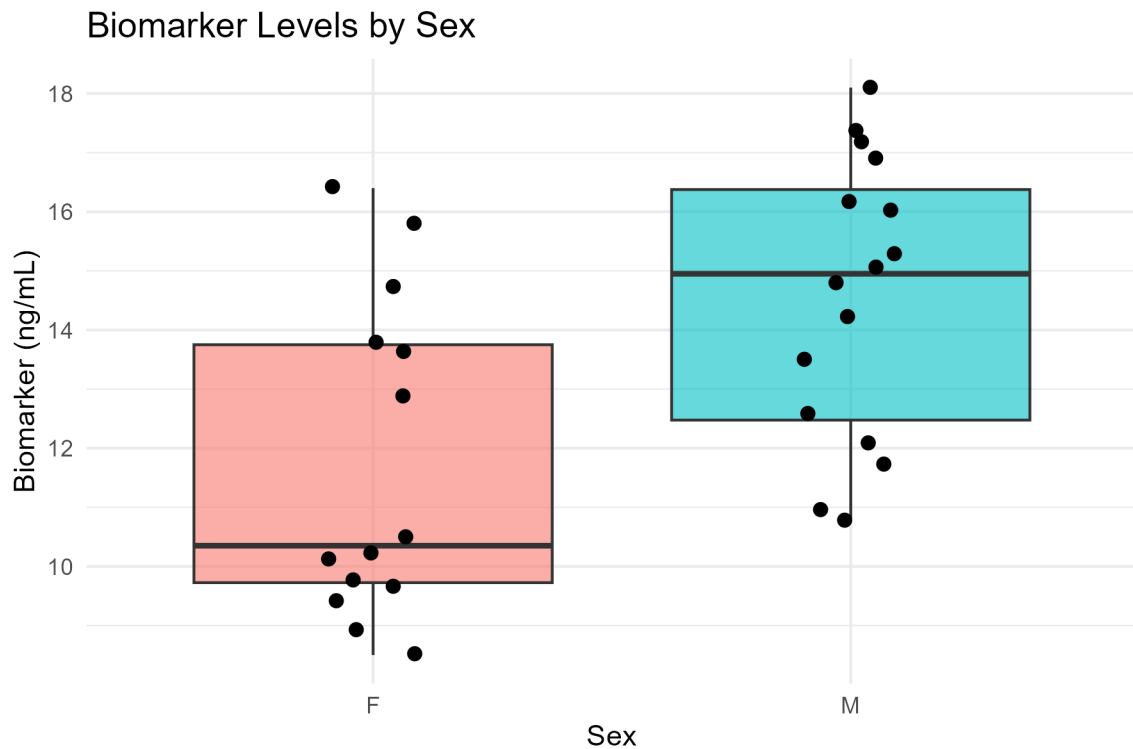


Figure 4: Biomarker Levels by Sex

5. Potential Confounding Factors

The Treatment group is systematically older than the Control group. Because age strongly elevates biomarker levels and is a confounding factor when assessing treatment efficacy.

6. Outliers or Unusual Observations

No numerical anomalies or extreme values in the dataset.

7. Additional Information Needed Before Drawing Conclusions

1. Longitudinal pre-treatment vs. post-treatment measurements.
 2. Equal distribution of age and sex across treatment arms.
 3. Additional clinical metadata like disease severity.
-

Reproducibility

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 22621)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United Kingdom.utf8
## [2] LC_CTYPE=English_United Kingdom.utf8
## [3] LC_MONETARY=English_United Kingdom.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United Kingdom.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] knitr_1.51      lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0
## [5] dplyr_1.2.1     purrr_1.2.2     readr_2.2.0    tidyr_1.3.2
## [9] tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      compiler_4.6.1    tidyselect_1.2.1  scales_1.4.0
## [5] yaml_2.3.12       fastmap_1.2.0     R6_2.6.1          generics_0.1.4
## [9] pillar_1.11.1     RColorBrewer_1.1-3 tzdb_0.5.0        rlang_1.3.0
## [13] stringi_1.8.7     xfun_0.60         S7_0.2.2          otel_0.2.0
## [17] timechange_0.4.0  cli_3.6.6         withr_3.0.3       magrittr_2.0.5
## [21] digest_0.6.39     grid_4.6.1        rstudioapi_0.19.0 hms_1.1.4
## [25] lifecycle_1.0.5   vctrs_0.7.3       evaluate_1.0.5    glue_1.8.1
## [29] farver_2.1.2      rmarkdown_2.31    tools_4.6.1       pkgconfig_2.0.3
## [33] htmltools_0.5.9
```